PCB Design Report

DIY SpO2 & Heart-Rate Monitor
Arduino Nano-Based Biosensor Interface with OLED Display

Submitted for: FOSSEE eSim PCB Design Task
Design Tool: eSim (KiCad-based schematic and PCB editor)
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1. Abstract

This report presents the design of a compact, Arduino Nano–based pulse oximeter and heart-rate monitoring device ("DIY SpO2"). The system uses a MAX30102 optical biosensor to measure blood-oxygen saturation (SpO2) and heart rate (BPM), an SSD1306 I2C OLED display for real-time readout, three push-buttons (Mode / Up / Down) for user interaction, and a status LED for alarm/activity indication. The complete design flow — schematic capture, component selection, bill of materials, pin mapping, PCB placement, and copper routing — was carried out in eSim (KiCad-based EDA tool) and is documented here along with the final routed board and its 3D rendered view. The design demonstrates key embedded-hardware concepts including I2C multi-device communication, digital input debouncing, mixed analog/digital PCB layout practices, and two-layer PCB routing.

2. Introduction

Pulse oximetry is a simple, non-invasive method of monitoring the oxygen saturation of a person's blood, along with heart rate, by shining light through (or reflecting light off) a peripheral part of the body such as a fingertip. With the increasing availability of low-cost integrated biosensor modules such as the MAX30102, it has become feasible for hobbyists, students, and embedded-systems learners to build their own pulse oximeter using an ordinary microcontroller board such as the Arduino Nano.

This project, referred to on the board silkscreen as "DIY SpO2", integrates the MAX30102 sensor, an SSD1306 OLED display, three tactile push-buttons, and a status LED around an Arduino Nano, and packages them onto a single custom PCB. The report covers the complete hardware design process, from the functional block diagram through to the final routed and rendered PCB, and is intended as project documentation for the FOSSEE eSim PCB Design Task.

3. Objectives

i. To design a schematic that interfaces a MAX30102 biosensor and an SSD1306 OLED display with an Arduino Nano over a shared I2C bus.

ii. To provide a simple three-button (Mode / Up / Down) user interface with hardware debouncing.

iii. To include a status/alarm LED for visual feedback.

iv. To lay out and route a compact, manufacturable 2-layer PCB in eSim, keeping the analog sensor isolated from digital switching noise.

v. To document the complete design — schematic, BOM, pin-mapping, and PCB routing — as a reference report.

4. System Block Diagram

At a system level, the board consists of five functional blocks connected to the Arduino Nano as shown below:

Block	Interface to Nano	Role
Power Supply (+5V/GND)	VIN / 5V / GND pins	Powers Nano, sensor, display, LED
MAX30102 Biosensor (U1)	I2C (A4/A5) + INT	Measures SpO2 and heart rate
SSD1306 OLED Display (U2)	I2C (A4/A5)	Displays readings / menu
Push-buttons (SW1-SW3)	Digital I/O (D2, D3, D4)	Mode select, Up, Down navigation
Status LED (D1)	Digital output (D12)	Alarm / activity indication

All peripheral communication (sensor + display) is multiplexed onto the same two-wire I2C bus, while the buttons and LED use dedicated single-purpose digital I/O lines.

5. Component Description

5.1 Arduino Nano (A1)

A compact ATmega328P-based development board acting as the system's main controller. It reads button inputs, communicates with the MAX30102 and SSD1306 over I2C, runs the SpO2/heart-rate calculation algorithm on the raw sensor samples, and drives the status LED and OLED output.

5.2 MAX30102 Biosensor (U1)

An integrated pulse-oximetry and heart-rate-monitor module containing red and IR LEDs, a photodetector, and onboard signal conditioning. It communicates over I2C (fixed address 0x57) and provides an interrupt (INT) output to notify the Arduino when new sample data is ready, avoiding the need for constant polling.

5.3 SSD1306 OLED Display (U2)

A 0.96-inch, 128x64 pixel monochrome OLED module driven over I2C (typical address 0x3C). It is used to display live BPM and SpO2 readings, along with any menu/mode information selected using the push-buttons.

5.4 Push-Buttons and Debounce Network (SW1-SW3, R1-R3, C1-C3)

Three tactile push-buttons, labelled on the silkscreen as "Down" (SW1), "Up" (SW2), and "Mode" (SW3), are each pulled up to +5V through a 1k resistor and filtered with a 100nF capacitor to ground. This RC network removes mechanical contact bounce so that the Arduino reads a single, clean logic transition per press.

5.5 Status LED (D1) and Current-Limiting Resistor (R4)

A single LED, labelled "alarm" on the silkscreen, is driven from Arduino pin D12 through a 1k current-limiting resistor (R4). It can be used to flag conditions such as low SpO2, sensor disconnection, or simply to indicate that a measurement is in progress.

6. Circuit Design and Working Principle

When the user places a finger on the MAX30102 sensor, its red and infrared LEDs illuminate the fingertip and the onboard photodetector measures the light reflected back. Because oxygenated and deoxygenated haemoglobin absorb red and infrared light differently, and blood volume in the fingertip pulses slightly with each heartbeat, the resulting waveform (the photoplethysmogram) can be processed to estimate both the heart rate (BPM) and blood-oxygen saturation (SpO2 %).

The MAX30102 buffers raw samples internally and raises its INT pin when a FIFO threshold is reached. The Arduino Nano services this interrupt, reads the samples over I2C, and runs a peak-detection / averaging algorithm to compute BPM and SpO2. These values are then written to the SSD1306 OLED over the same I2C bus for the user to read.

The three push-buttons let the user navigate a simple on-screen menu: SW3 ("Mode") cycles between display modes (e.g., live reading vs. history), while SW2 ("Up") and SW1 ("Down") scroll or adjust values within a mode. The status LED (D1) gives an at-a-glance indication, such as blinking with each detected pulse or lighting solidly for an alarm condition (e.g., SpO2 below a safe threshold).

Because U1 (MAX30102) and U2 (SSD1306) share the same SDA/SCL lines, both devices must use distinct I2C addresses. This is satisfied by design, as the MAX30102 is fixed at 0x57 and the SSD1306 module used here defaults to 0x3C — no address conflict or I2C multiplexer is required.

7. Schematic Diagram

Figure 1 shows the complete schematic capture of the design, produced in eSim.

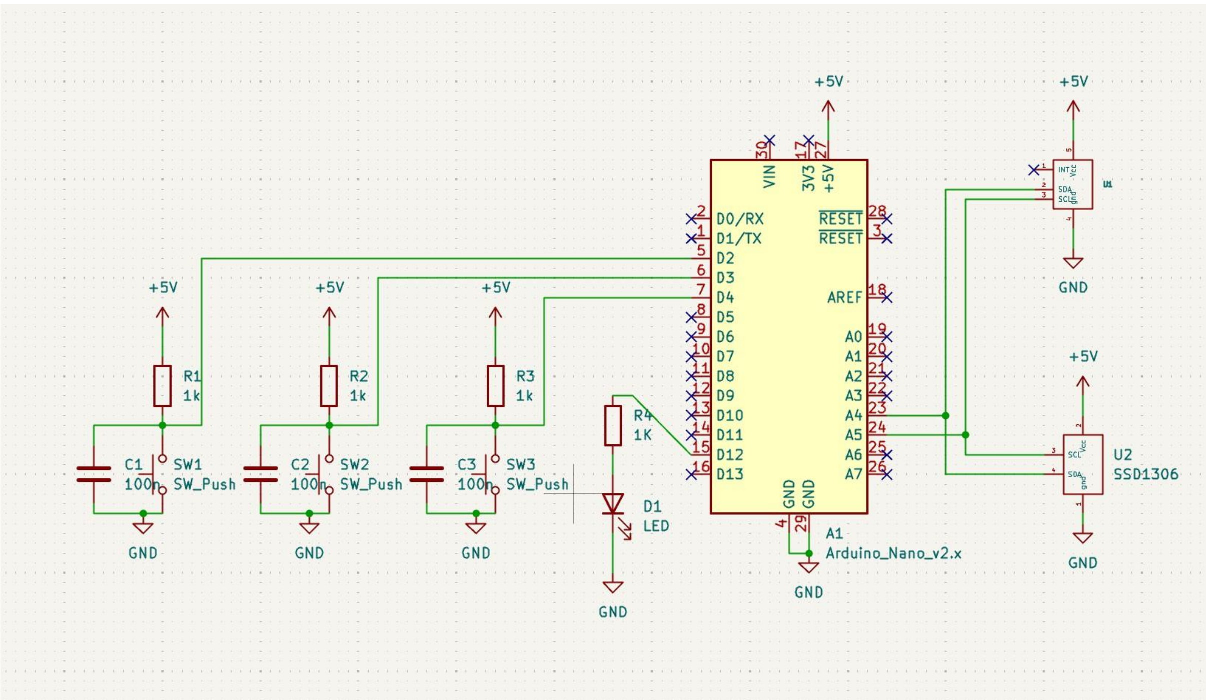

Figure 1: Full circuit schematic (Arduino Nano, MAX30102, SSD1306, buttons, LED).

8. Bill of Materials (BOM)

Ref. Des.	Component	Value / Part	Qty	Function
A1	Microcontroller module	Arduino Nano v2.x	1	Main controller
U1	Biosensor module	MAX30102	1	HR / SpO2 sensing (I2C)
U2	OLED display	SSD1306, 0.96" I2C	1	Data / menu display
SW1-SW3	Tactile push-button	SW_Push	3	User input (Down/Up/Mode)
R1-R3	Resistor	1k ohm	3	Button pull-up
R4	Resistor	1k ohm	1	LED current limit

D1	LED	5mm, generic	1	Alarm / status indicator
C1-C3	Ceramic capacitor	100 nF	3	Button debounce filter

9. Arduino Pin Mapping

Signal	Arduino Pin	Connected To
Button - Down	D2	SW1 (pull-up + debounce)
Button - Up	D3	SW2 (pull-up + debounce)
Button - Mode	D4	SW3 (pull-up + debounce)
LED drive (alarm)	D12	D1 via R4
I2C data	A4 (SDA)	MAX30102 & SSD1306
I2C clock	A5 (SCL)	MAX30102 & SSD1306
Power	5V / GND	All modules

10. PCB Layout and Routing

The board was placed and routed in eSim. Figure 2 shows the top-view copper routing (top layer in red, bottom layer in blue) and Figure 3 shows the 3D rendered view of the completed, assembled board.

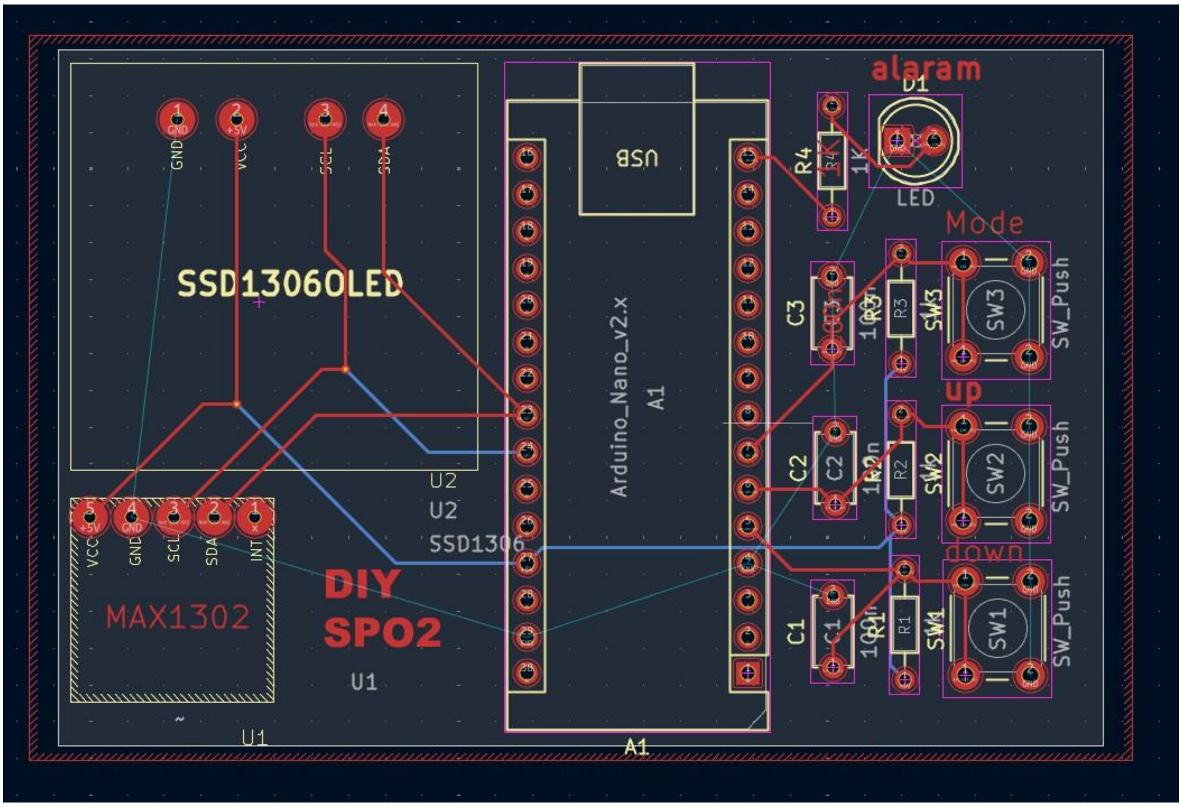

Figure2:Top-viewPCB routing layout (red = top copper, blue = bottom copper).

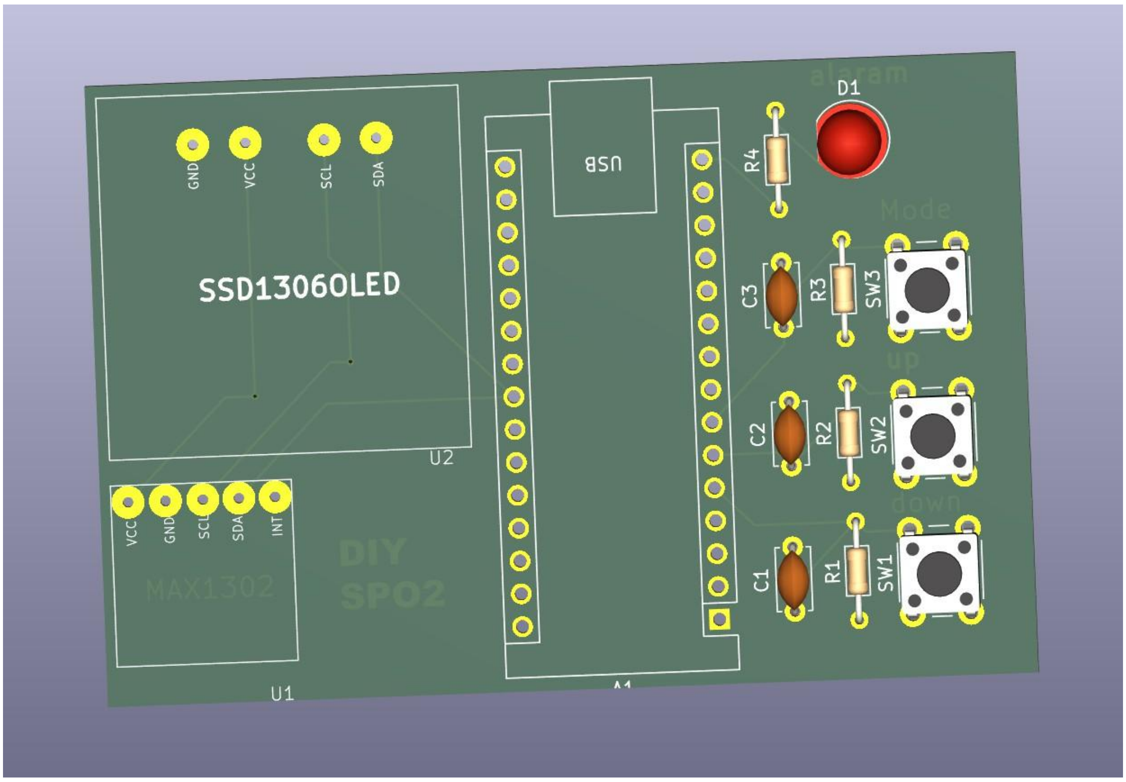

Figure 3: 3D rendered view of the assembled PCB.

10.1 Board Specifications

Layers: 2-layer PCB — top copper (red) and bottom copper (blue).

Board outline: rectangular, approx. 85mm x 55mm (estimated from layout proportions), edge-cut border shown by the red/white hatched outline.

Arduino Nano (A1) mounted via two long female headers running down the centre of the board, USB port oriented toward the top edge for easy programming access.

U2 (SSD1306) and U1 (MAX30102) connect via 4-pin and 5-pin headers respectively, on the left half of the board.

Silkscreen labels: "Mode" (SW3), "Up" (SW2), "Down" (SW1), "alarm" (D1/LED), and the project label "DIY SPO2".

10.2 Component Placement

Arduino Nano (A1) placed centrally, spanning the full height of the board, with header pins broken out on both sides for routing access.

U2 (SSD1306 OLED) placed top-left, isolated from the button/LED cluster for clear user visibility; GND/VCC/SCL/SDA pins routed directly to the Nano's I2C pins.

U1 (MAX30102) placed bottom-left, below the OLED, keeping the optical sensor away from the digital switching noise of the buttons and LED on the right side of the board.

Right-hand cluster: D1 (alarm LED) and R4 at the top, followed by three RC-debounced button groups (C3/R3/SW3 "Mode", C2/R2/SW2 "Up", C1/R1/SW1 "Down") stacked vertically, each within its own placement/courtyard outline.

USB connector footprint placed top-centre, above the Nano, keeping the programming port accessible at the board edge.

10.3 Routing Details

I2C bus (SDA/SCL): routed from the Nano's A4/A5 pins to both U1 and U2 using a mix of top (red) and bottom (blue) layer traces, with layer transitions at via points near the Nano header — visible as the crossing red/blue diagonal traces between the two modules.

Power distribution: +5V and GND (red traces) run from the Nano's power pins to the OLED and MAX30102 headers along the left side of the board.

Button and LED signal traces: red top-layer traces connect each SW/R/C debounce network to the corresponding Nano digital pin (D2/D3/D4) and D12 for the LED, kept short and direct within the right-hand cluster.

Thin cyan lines in the layout view are net/ratsnest reference guides for placement and do not represent unrouted copper.

Analog (MAX30102) and digital-switching (button/LED) traces are kept physically separated by routing them on opposite sides of the Arduino Nano footprint, reducing coupling between the sensitive optical front-end and noisy digital signals.

11. Design Verification Checklist

Item	Status
Schematic ERC (electrical rules check) passed	To be verified in eSim
All components assigned footprints	Complete
I2C addresses confirmed non-conflicting (0x57, 0x3C)	Confirmed by design
Component placement finalized	Complete
PCB routing (top + bottom copper) completed	Complete
DRC (design rule check) passed on PCB layout	To be verified in eSim
Gerber files generated for fabrication	Pending export from eSim

12. Applications

Personal, low-cost heart-rate and SpO2 self-monitoring device.

Educational demonstration of biosensor interfacing and I2C multi-device communication.

Base platform for wearable or bedside health-monitoring prototypes.

Reference design for students learning embedded systems and PCB design workflows.

13. Advantages and Limitations

13.1 Advantages

Low component count and low cost compared to commercial pulse oximeters.

Simple three-button interface requiring no external computer for operation.

Standard I2C interfacing makes the design easy to extend with additional sensors.

Compact 2-layer PCB keeps manufacturing simple and low-cost.

13.2 Limitations

Not a certified medical device; readings are indicative only and should not be used for clinical diagnosis.

Measurement accuracy is sensitive to finger placement, motion artefacts, and ambient light on the MAX30102 sensor.

No onboard battery management circuitry in the current design; the board is powered externally via +5V.

14. Future Scope

Add a battery and charging/power-management circuit for portable, wearable use.

Add Bluetooth/Wi-Fi connectivity (e.g., an ESP32 variant) for logging readings to a phone app.

Add data-logging to onboard flash/SD card for long-term trend tracking.

Enclose the sensor in a finger-clip housing to improve measurement repeatability.

15. Conclusion

This report has documented the complete design of a DIY SpO2 and heart-rate monitoring board built around an Arduino Nano, a MAX30102 biosensor, an SSD1306 OLED display, three push-buttons, and a status LED. The schematic, bill of materials, pin mapping, and final PCB layout and routing have all been presented. The design achieves its objectives of providing a functional, low-cost, and manufacturable pulse-oximeter board while demonstrating good PCB design practice, including I2C bus sharing, input debouncing, and separation of analog sensor circuitry from digital switching noise. The project is well suited to the FOSSEE eSim PCB Design Task as it exercises schematic capture, component placement, and multi-layer routing within a single, moderately-scoped, real-world design.